

TABLES DI 12-9-26

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

Percentage of students at five business schools who opted for different specializations.

Business school → Specialization ↓	P	Q	R	S	T
Finance	24	25	18	12	10
HR	12	15	16	24	25
Operations	18	20	16	24	20
Marketing	36	15	20	28	10
Systems	10	25	30	12	35

Note: The percentages are the same for male and female students.

Ratio of the number of male and female students in the five business schools and the total number of students in each business school is given in the following table.

Business	Male : Female	Total
P	3 : 2	250
Q	5 : 3	320
R	4 : 3	350
S	4 : 5	225
T	7 : 5	480

Directions for question 1: Type in your answer in the input box provided below the question.

How many students opted for marketing in school S?

Your Answer: 63 ✓

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Solution:

The given information can be tabulated as below.

Specialization / school	P		Q		R		S		T	
	Males	Females	Males	Females	Males	Females	Males	Females	Males	Females
Finance	36	24	50	30	36	27	12	15	28	20
HR	18	12	30	18	32	24	24	30	70	50
Operations	27	18	40	24	32	24	24	30	56	40
Marketing	54	36	30	18	40	30	28	35	28	20
Systems	15	10	50	30	60	45	12	15	98	70
Total	150	100	200	120	200	150	100	125	280	200

Number of students opting marketing from school S is $28 + 35 = 63$.

Ans : (63)

Directions for question 2: Select the correct alternative from the given choices.

What is the ratio of the number of female students in school S to that of male students in school T?

- ☐ a) 5 : 6
- ☐ b) 15 : 28
- ☐ c) 4 : 7

☒ d) 25 : 56 (Your answer - ✓)

Directions for questions 3 and 4: Type in your answer in the input box provided below the question.

What is the approximate percentage of students who opted for systems in all the five schools together?

Directions for questions 3 and 4: Type in your answer in the input box provided below the question.

How many students opted for operations in school R?

Your Answer: 56 ✓

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Solution:

The given information can be tabulated as below.

Specialization / school	P		Q		R		S		T	
	Males	Females	Males	Females	Males	Females	Males	Females	Males	Females
Finance	36	24	50	30	36	27	12	15	28	20
HR	18	12	30	18	32	24	24	30	70	50
Operations	27	18	40	24	32	24	24	30	56	40
Marketing	54	36	30	18	40	30	28	35	28	20
Systems	15	10	50	30	60	45	12	15	98	70
Total	150	100	200	120	200	150	100	125	280	200

Number of students who opted for operations in school R = 32 + 24 i.e., 56.

Ans : (56)

Directions for question 5: Select the correct alternative from the given choices.

What is the ratio of the number of male students who opted for HR in school P and the number of female students who opted for Finance in school Q?

- ☐ a) 3 : 5
- ☐ b) 25 : 16
- ☐ c) 15 : 26
- ☐ d) 5 : 3

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Solution:

The given information can be tabulated as below.

Specialization / school	P		Q		R		S		T	
	Males	Females	Males	Females	Males	Females	Males	Females	Males	Females
Finance	36	24	50	30	36	27	12	15	28	20
HR	18	12	30	18	32	24	24	30	70	50
Operations	27	18	40	24	32	24	24	30	56	40
Marketing	54	36	30	18	40	30	28	35	28	20
Systems	15	10	50	30	60	45	12	15	98	70
Total	150	100	200	120	200	150	100	125	280	200

Required ratio = 18 : 30 i.e., 3 : 5.

Choice (A)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

The table gives data about the weights of students in five schools.

School	Classification based on the weight of the student				
	Above 15 kg but below 25 kg	Above 15 kg but below 35 kg	Above 15 kg but below 45 kg	Atleast 45 kg but below 55 kg	55 kg or more
Green Lands Techno School	110	135	146	42	16
The Adventist High School	164	195	212	56	12
Rainbow Public School	98	106	162	38	18
Little Flower Global School	48	128	198	64	21
Blue Birds International School	85	138	152	72	33

Note: No student weighs 15 kg or less.

In which of the following schools is the ratio of the number of students whose weight is more than 15 kg but below 45 kg to the total number of students in that school, the highest?

- ☐ a) Green Lands Techno School
- ☐ b) The Adventist High School
- ☐ c) Rainbow Public School
- ☐ d) Little Flower Global School

Solution:

Required ratio for given school is as follows.

Green Lands Techno School = $146/204$

$$= \left(1 - \frac{58}{204}\right) = \left(1 - \frac{1}{4}\right)$$

The Adventist High School = $212/280$

$$= \left(1 - \frac{68}{280}\right) = 1 - \frac{1}{4}$$

Rainbow Public School = $162/218$

$$= \left(1 - \frac{56}{218}\right) = \left(1 - \frac{1}{4}\right)$$

Little Flower Global School = $198/283$

$$= \left(1 - \frac{85}{283}\right) = \left(1 - \frac{1}{4}\right)$$

Blue Birds International School = $152/257$

$$= 1 - \frac{105}{257} = 1 - \frac{1}{3}$$

Therefore Adventist High school has the highest.

Choice (B)

In which of the following groups is the number of students of all the given schools together the highest?

- ☐ a) Below 25 kg
- ☐ b) At least 25 kg but below 45 kg
- ☐ c) At least 35 kg but below 55 kg
- ☐ d) At least 25 kg but below 55 kg

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Solution:

The total number of students whose weight is above 15 kg but below 25 kg in all the schools = 505
Those whose weight is 25 kg or above but below 35 kg
= $702 - 505 = 197$
Those weighing above 35 kg but below 45 kg = $870 - 702 = 168$
Those weighing above 45 kg but below 55 kg = 272
Those weighing above 55 kg = 100.

Choice (D)

If 35 students, each of whose weight is below 30 kg, join Rainbow Public School, what would be the approximate percentage of the total students weighing below 35 kg?

- ☐ a) 56
- ☐ b) 48
- ☐ c) 52
- ☐ d) 61

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Solution:

Total number of students whose weight is below 35 kg but above 15 kg = $106 + 35 = 141$
Total number of students in Rainbow Public school = 253
Required percentage = $\left(\frac{141}{253}\right) \times 100\% \approx 56\%$.

Choice (A)

Which of the following could be the average weight of the students, of Little Flower Global School, whose weight is below 55 kg?

- ☐ a) 28
- ☐ b) 35
- ☐ c) 42
- ☐ d) 49

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Solution:

The required average weight of the students in Little Flower Global School is $> (48 \times 15 + 80 \times 25 + 70 \times 35 + 64 \times 45)/262 = 8050/262 \approx 30.7$ kg.
The required average weight of the students in Little Flower Global School is $< (48 \times 25 + 80 \times 35 + 70 \times 45 + 64 \times 55)/262 = 10670/262 \approx 40.7$ kg.
Only 35 lies between the two values.

Choice (B)

What percentage of the number of students in the school that has the highest number of students whose weight is above 15 kg but below 55 kg is the number of students in the school that has the lowest number of students whose weight is atleast 25 kg?

- ☐ a) 56.71
- ☐ b) 63.14
- ☐ c) 82.57
- ☐ d) 72.85

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Solution:

The school with the highest number of students whose weight is above 15 kg but below 55 kg is The Adventist High School with 268 students.
The school that has the lowest number of students whose weight is above 25 kg is Green Lands Techno School and it has 204 students satisfying this condition.
Required percentage = $(204/268) \times 100 = 72.85\%$.

Choice (D)

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

Production (in thousand tonnes) of Rice, Oil seeds and Sugar Cane by six states in the year 2011-12.

States	Rice	Oil Seeds	Sugar Cane
Andhra Pradesh	12881	13288	39%
Maharashtra	2805	28%	8859
Gujarat	10%	4920	12000
West Bengal	89%	744	1170
Uttar Pradesh	14025	885	65%
Punjab	10530	5%	4670

Note: The data is regarding only these three crops. 39% of the total production of Andhra Pradesh is sugarcane.

What is the ratio of the production of rice in Gujarat to the production of oil seeds in Punjab?

- ☐ a) 37 : 49
- ☐ b) 20 : 47
- ☐ c) 40 : 27
- ☐ d) None of these

Solution:

States	Rice	Oil Seeds	Sugar Cane
A.P.	12881	13288	$\frac{26169 \times 39}{61} = 16731$
Maharashtra	2805	$\frac{11664 \times 28}{72} = 4536$	8859
Gujarat	$\frac{16920}{90} \times 10 = 1880$	4920	12000
West Bengal	$\frac{1914}{11} \times 89 = 15486$	744	1170
Uttar Pradesh	14025	885	$\frac{14910}{35} \times 65 = 27690$
Punjab	10530	$\frac{15200 \times 5}{95} = 800$	4670

Production of Rice in the Gujarat = 1880

Production of Oil Seeds in the Punjab = 800

∴ Required ratio = 1880 : 800 = 47 : 20.

Choice (D)

Quantity of oil seeds produced in Maharashtra is approximately what percent of the total oilseeds production in all the six states?

- ☐ a) 18%
- ☐ b) 24%
- ☐ c) 32%
- ☐ d) 16%

Solution:

States	Rice	Oil Seeds	Sugar Cane
A.P.	12881	13288	$\frac{26169 \times 39}{61} = 16731$
Maharashtra	2805	$\frac{11664 \times 28}{72} = 4536$	8859
Gujarat	$\frac{16920}{90} \times 10 = 1880$	4920	12000
West Bengal	$\frac{1914}{11} \times 89 = 15486$	744	1170
Uttar Pradesh	14025	885	$\frac{14910}{35} \times 65 = 27690$
Punjab	10530	$\frac{15200 \times 5}{95} = 800$	4670

Required Percentage

$$= \left(\frac{4536}{13288 + 4536 + 4920 + 744 + 885 + 800} \right) \times 100$$

$$= \frac{4536}{25173} \times 100 = 18.01 \approx 18\%.$$

Choice (A)

What is the average production (in thousand tonnes) of sugar cane in the given six states?

- ☐ a) 81531.36
- ☐ b) 12643.99
- ☐ c) 141562.222
- ☐ d) 11853.333

Solution:

States	Rice	Oil Seeds	Sugar Cane
A.P.	12881	13288	$\frac{26169 \times 39}{61} = 16731$
Maharashtra	2805	$\frac{11664 \times 28}{72} = 4536$	8859
Gujarat	$\frac{16920}{90} \times 10 = 1880$	4920	12000
West Bengal	$\frac{1914}{11} \times 89 = 15486$	744	1170
Uttar Pradesh	14025	885	$\frac{14910}{35} \times 65 = 27690$
Punjab	10530	$\frac{15200 \times 5}{95} = 800$	4670

Required average
$$= \frac{(16731 + 8859 + 12000 + 1170 + 27690 + 4670)}{6}$$
$$= \frac{71120}{6} = 11853.333 \text{ (thousand tones).}$$

Choice (D)

What is the total production of rice (in thousand tonnes) in the given six states?

- ☐ a) 36507
- ☐ b) 57607
- ☐ c) 67507
- ☐ d) 75607

Solution:

States	Rice	Oil Seeds	Sugar Cane
A.P.	12881	13288	$\frac{26169 \times 39}{61} = 16731$
Maharashtra	2805	$\frac{11664 \times 28}{72} = 4536$	8859
Gujarat	$\frac{16920}{90} \times 10 = 1880$	4920	12000
West Bengal	$\frac{1914}{11} \times 89 = 15486$	744	1170
Uttar Pradesh	14025	885	$\frac{14910}{35} \times 65 = 27690$
Punjab	10530	$\frac{15200 \times 5}{95} = 800$	4670

Total Rice production
= 12881 + 2865 + 1880 + 15486 + 14025 + 10530
= 57607 (thousand tones).

Choice (B)

Sugar cane produced in Andhra Pradesh is approximately what percent of that produced in Uttar Pradesh?

- ☐ a) 60%
- ☐ b) 40%
- ☐ c) 50%
- ☐ d) 55%

Solution:

States	Rice	Oil Seeds	Sugar Cane
A.P.	12881	13288	$\frac{26169 \times 39}{61} = 16731$
Maharashtra	2805	$\frac{11664 \times 28}{72} = 4536$	8859
Gujarat	$\frac{16920}{90} \times 10 = 1880$	4920	12000
West Bengal	$\frac{1914}{11} \times 89 = 15486$	744	1170
Uttar Pradesh	14025	885	$\frac{14910}{35} \times 65 = 27690$
Punjab	10530	$\frac{15200 \times 5}{95} = 800$	4670

Required percentage = $\frac{16731}{27690} \times 100 = 60.422 \approx 60\%$.

Choice (A)

